Dense Retrieval Basics

1 Introduction.

Dense retrieval is a method that finds passages by comparing embedding vectors of the query and the passages. A bi-encoder is a model that encodes the query and each passage separately into vectors. Cosine similarity is a measure of the angle between two embedding vectors.

2 Strengths and costs.

Dense retrieval captures meaning that shares no words with the query. Dense retrieval needs an index of vectors built ahead of time. Approximate nearest neighbour search makes dense retrieval fast on large collections. Dense retrieval can miss rare names and acronyms that a keyword method finds easily.

3 Training and indexing.

A bi-encoder is trained on pairs of questions and the passages that answer them. Dense retrieval depends on the quality of the embedding vectors a bi-encoder produces. Approximate nearest neighbour search trades a little accuracy for a large gain in speed. Cosine similarity is the usual score for comparing the embedding vectors of a query and a passage. Hybrid retrieval adds a keyword method to dense retrieval so that rare names are not lost.

4 Limits.

Dense retrieval can return passages that sound related but do not answer the question. A reranker can correct the order of the passages that dense retrieval returns. Dense retrieval and BM25 fail in different ways, which is why hybrid retrieval combines them.